Max Bregler

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Education

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. Candidate in Mathematics of Computation (GPA: 3.83/4.0) — *Expected Graduation: June 2029*

Professional Experience

AugX Labs — New York, NY
Intern — *June 2023 – August 2023*
- Developed for company's flagship product (AI-powered web-based video editing software) using JS
- Developed transition system using FFMpeg (PHP backend) and Remotion (React JS frontend)
- Researched potential solution to automatically turn AirBnB listings into video advertisements

Pinscreen Inc. — Los Angeles, CA
Research Intern — *June 2022 – November 2022*
- Developed Unity (C#) demo for company's flagship product (AI-generated 3D avatars) integrating webcam
- Researched and implemented image upscaling (GANs) for restoring materials for Balenciaga campaign

Projects

VisitNU — San Francisco, CA
Project Lead — *January 2025 – May 2025*
- Revamped high school's campus security system to allow RFID tags to check in and out of school
- Integrated RFID tags (C & Python for backend) into VisitU (Bay Area campus security company)
- Acquired 150 student signups in 2 days, garnering over 1000 uses per week of school with positive feedback

Vids2Model — San Francisco, CA
Project Lead — *December 2023 – May 2024*
- Developed CLI tool (TensorFlow) to create image classification models from video input without expertise
- Created website (PHP) with user queue to implement CLI tool with intuitive design

Extracurriculars

Creative Labs @ UCLA — Los Angeles, CA
Full-Stack Developer (TreeReq) — *April 2026 – Present*
- Developed visual course planner for UCLA students alongside several other developers
- Implemented authentication and data management systems (Python, MongoDB)
- Integrated systems with frontend (FastAPI) and improved onboarding UI (ReactJS)

BruinML — Los Angeles, CA
Researcher (Emergent Intelligence) — *January 2026 – Present*
- Explored mathematical principles of Multi Arm Bandits using calculus, proving well-known RL theorems
- Discussed viability of implementing stigmergy to bandit problems to decrease environmental effects of ML

ACM Studio @ UCLA — Los Angeles, CA
Student Run Studios Developer (Spritide) — *September 2025 – Present*
- Developed topdown roguelike action RPG (Unity) with complex combat system with several other developers
- Implemented performance increases for game like object pooling hitboxes and hitscans
- Implemented core features in combat loop like critical hits and defense calculations

AI Safety @ UCLA — Los Angeles, CA
Technical Fellow (Reinforcement Learning) — *September 2025 – December 2025*
- Explored mathematical principles of MDPs using linear algebra and multivariable calculus from basics
- Developed MDP policies for many environments, including Lunar Lander and Cartpole
- Discussed academic papers from both universities and AI labs on existential risks of AI (scaling laws, etc)

Skills & Coursework

Languages: C, C++, C#, Python, JavaScript, PHP
Libraries & Tools: TensorFlow, ReactJS, FastAPI, Remotion, MongoDB, Unity, FFMpeg, Git
Coursework: MATH 115A (Abstract Linear Algebra) 131A (Real Analysis) 151A (Numerical Analysis)